

Software Requirements Specification

Project Pitch Perfect

Formal SRS and Implementation Planning Package

Field	Value
Prepared for	Douglas J. Sweeting, Sr.
Version	1.0
Date	June 7, 2026
Status	Draft for architecture, development, and Codex-assisted implementation
Primary goal	Build a music training platform for pitch, rhythm, practice, instruction, analytics, and institutional use

Version	Date	Author	Description
1.0	June 7, 2026	ChatGPT for Douglas J. Sweeting, Sr.	Initial consolidated SRS based on previously supplied requirements.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, nonfunctional, architectural, data, interface, security, testing, and deployment requirements for Project Pitch Perfect. The document is intended to guide AI-assisted code generation, developer implementation, testing, procurement discussions, school or institutional review, and future product planning.

1.2 Product Scope

Project Pitch Perfect is a music education and performance training application that provides real-time pitch detection, note recognition, rhythm and timing feedback, metronome support, audio recording, progress tracking, role-based portals, reporting, cloud synchronization, remote learning support, and institution-level administration.

1.3 Intended Audience

The intended audience includes the product owner, software developers, AI coding assistants, testers, UI designers, cloud engineers, school administrators, music teachers, parents, and future investors or partners.

1.4 Product Vision

The product vision is to help students practice voice or instruments with accurate, measurable feedback while giving teachers, parents, and institutions visibility into progress, consistency, skill development, and instructional needs.

2. Overall Description

2.1 Product Perspective

The system shall operate as a multi-platform music training ecosystem. It may begin as a desktop or web prototype and evolve into a cloud-supported platform with browser, mobile, and desktop clients.

2.2 Product Functions

Core product functions include practice sessions, target note exercises, pitch detection, metronome-guided rhythm training, recording, playback, scoring, analytics, assignments, class management, teacher feedback, parent access, administration, and reporting.

2.3 User Classes

Primary user classes are students, teachers, parents, institution administrators, and system administrators. Future user classes may include school district administrators, music program directors, support staff, and external evaluators.

2.4 Operating Environment

The system should support modern web browsers and may support Windows, macOS, Linux, Android, and iOS through native, hybrid, or progressive web application deployments.

3. Stakeholders and User Roles

Role	Primary Responsibilities	Primary Permissions	Restrictions
Student	Practice, complete assignments, record sessions, view progress, ask questions	Practice tools, personal dashboard, assigned lessons, own recordings	Cannot view other students or instructor-only rankings by default
Teacher	Manage classes, assign exercises, review performance, communicate with students	Class creation, student rosters, assignments, analytics, reports	Cannot access unrelated institutional data unless granted
Parent	Monitor assigned student progress and practice compliance	View linked student reports, achievements, notifications	Cannot modify grades, assignments, or teacher records
Institution Administrator	Manage teachers, classes, departments, subscriptions, institutional reporting	Manage institutional users, licenses, reports, policies	Cannot access system-wide controls outside institution
System Administrator	Operate and configure platform	Full platform administration, support, licensing, content configuration	Subject to audit logging and least-privilege operational controls

4. Authentication and Account Management

ID	Requirement	Priority	Acceptance Criterion
AUTH-001	The system shall require a unique account for each user.	Must	A user cannot register with a duplicate email address.
AUTH-002	The system shall support email and password authentication.	Must	A registered user can log in with verified credentials.
AUTH-003	The system should support Google OAuth and Microsoft OAuth.	Should	A user can authenticate through approved identity providers.
AUTH-004	The system should support institutional single sign-on.	Should	A school can configure SSO for authorized users.
AUTH-005	Passwords shall never be stored in plain text.	Must	Password hashes are stored using an industry-

			standard algorithm.
AUTH-006	The system shall support role-based access control.	Must	Student, teacher, parent, administrator, and system administrator permissions are enforced.

5. Student Practice Module

ID	Requirement	Priority	Acceptance Criterion
STU-001	The system shall allow students to start a practice session.	Must	A student can select an exercise and begin recording or live monitoring.
STU-002	The system shall show the target note, detected note, frequency, and accuracy feedback.	Must	The practice screen updates during live input.
STU-003	The system shall support voice and instrument practice workflows.	Must	A student can choose voice or instrument mode before practice.
STU-004	The system shall track practice duration.	Must	Session duration is stored with the student record.
STU-005	The system shall support a daily practice target of 15 minutes by default.	Must	The dashboard displays daily progress toward 15 minutes.
STU-006	Classroom instruction time shall not count toward daily practice by default.	Should	Teacher-led class time is stored separately unless policy changes it.

6. Pitch Detection and Audio Analysis

ID	Requirement	Priority	Acceptance Criterion
AUD-001	The system shall detect pitch from microphone audio input in real time.	Must	The detected frequency and note are displayed with acceptable latency.
AUD-002	The system shall recognize natural notes A through G, sharps, and flats.	Must	The system maps detected frequency to the nearest note.
AUD-003	The system shall calculate deviation from target pitch.	Must	The system reports flat, sharp, or on-pitch status.
AUD-004	The system should support noise reduction.	Should	Background noise is reduced before pitch analysis where supported.
AUD-005	The system should detect whether the input is likely voice or instrument.	Should	The system labels or adjusts analysis mode based on input characteristics.
AUD-006	The system shall allow calibration for microphone	Must	Users can test input levels before practice.

	sensitivity.		
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7. Rhythm and Metronome Module

ID	Requirement	Priority	Acceptance Criterion
MET-001	The system shall provide an adjustable metronome.	Must	A user can select BPM and start or stop the metronome.
MET-002	The system shall support multiple common time signatures.	Must	The metronome can operate in 2/4, 3/4, 4/4, and 6/8.
MET-003	The system shall provide both audio and visual beat indicators.	Must	Beat feedback is visible and audible.
MET-004	The system shall measure tempo adherence during assigned exercises.	Should	Reports include timing consistency and beat accuracy.

8. Recording and Playback

ID	Requirement	Priority	Acceptance Criterion
REC-001	The system shall allow students to record practice sessions.	Must	A recorded session can be saved and replayed.
REC-002	The system shall store recording metadata.	Must	Date, time, duration, exercise, instrument mode, and user are stored.
REC-003	The system should support cloud storage for recordings.	Should	Recordings synchronize to the user account when online.
REC-004	The system should allow students and teachers to compare recordings over time.	Should	Two or more recordings can be selected for review.

9. Teacher Portal

ID	Requirement	Priority	Acceptance Criterion
TCH-001	The system shall allow teachers to create and manage classes.	Must	A teacher can create a class and add students.
TCH-002	The system shall support up to 30 active students per class as a baseline requirement.	Must	A class roster can contain 30 students without functional failure.
TCH-003	Teachers shall be able to create assignments.	Must	Students can see and complete assigned exercises.

TCH-004	Teachers shall be able to view student analytics.	Must	Teacher dashboard shows pitch, timing, practice, and progress data.
TCH-005	Teachers shall be able to respond to student questions by chat or email.	Should	A student question can be answered through configured communication channels.
TCH-006	Teachers shall be able to generate class and student reports.	Must	PDF and CSV exports are available.

10. Parent Portal

ID	Requirement	Priority	Acceptance Criterion
PAR-001	The system shall allow parents to view linked student progress.	Must	A parent account can access only approved student records.
PAR-002	The system shall show practice compliance and achievements.	Must	Parent dashboard displays practice history and milestones.
PAR-003	The system should send parent notifications.	Should	Parents can receive alerts for missed practice or achievements.

11. Institution Administration

ID	Requirement	Priority	Acceptance Criterion
ADM-001	Institution administrators shall manage teachers and classes.	Must	An administrator can create, edit, or deactivate institutional users.
ADM-002	Institution administrators shall manage subscriptions and licenses.	Must	The institution dashboard displays seats, usage, and license status.
ADM-003	Institution administrators shall generate institution-wide reports.	Must	Reports can summarize usage and progress across classes.
ADM-004	The system should support departments or programs.	Should	Classes and teachers can be grouped by department, grade, school, or program.

12. AI Laboratory and Feedback Controls

ID	Requirement	Priority	Acceptance Criterion
AI-001	The system may provide AI-generated feedback on student practice.	Could	AI feedback appears when the feature is enabled.
AI-002	The AI Lab shall show a visible AI ON/OFF	Must	Users can clearly see whether AI analysis is

	indicator.		enabled.
AI-003	Instructor control shall be required for AI features in class settings.	Must	Students cannot independently enable restricted AI classroom features.
AI-004	Student performance data shall not be used for AI training unless explicitly enabled by authorized policy.	Must	Data usage settings are stored and enforceable.

13. Reporting and Analytics

ID	Requirement	Priority	Acceptance Criterion
RPT-001	The system shall calculate pitch accuracy, timing accuracy, practice frequency, duration, and improvement trends.	Must	Metrics are stored and displayed on dashboards.
RPT-002	The system shall export reports in PDF and CSV formats.	Must	User can download selected reports.
RPT-003	Reports shall be role-specific.	Must	Students, parents, teachers, and administrators see only authorized data.
RPT-004	The system should support longitudinal progress views.	Should	Reports can show weekly, monthly, semester, and custom date ranges.

14. Profiles, Avatars, and Achievements

ID	Requirement	Priority	Acceptance Criterion
PRO-001	The default student profile shall include student name.	Must	A student record can be created with name only.
PRO-002	Optional student fields shall include student ID, grade, and school.	Should	Teachers or administrators can add optional fields.
PRO-003	Teachers shall be able to create student avatars.	Should	A teacher can assign or customize an avatar.
PRO-004	Default avatars shall be racially neutral and non-descriptive.	Must	Default avatar library avoids racialized or sensitive assumptions.
ACH-001	The system should support achievements and milestones.	Should	Students and parents can view earned achievements.

15. Anti-Bullying and Student Safety

ID	Requirement	Priority	Acceptance Criterion
SAFE-001	The system shall not expose student ranking lists publicly to other students.	Must	Student dashboards do not show other student rankings.
SAFE-002	Instructor-only rankings shall be restricted.	Must	Beginner, Novice, Competent, and Expert classifications are visible only to instructors unless enabled.
SAFE-003	The system should support respectful feedback wording.	Should	Feedback is instructional rather than humiliating or punitive.

16. Nonfunctional Requirements

ID	Requirement	Priority	Acceptance Criterion
NFR-001	Live pitch feedback should feel responsive to the user.	Must	Target latency is low enough for practical practice; exact threshold to be validated in prototype.
NFR-002	The system shall protect student data through encryption in transit.	Must	All web traffic uses HTTPS/TLS.
NFR-003	Sensitive stored data shall be encrypted or protected according to platform best practices.	Must	Credentials, tokens, and protected records are not stored in plain text.
NFR-004	The system shall maintain audit logs for administrative actions.	Must	Creation, deletion, role changes, and access changes are logged.
NFR-005	The system shall support accessibility best practices.	Must	Core screens are keyboard navigable and readable by assistive technologies.
NFR-006	The system shall degrade gracefully when offline.	Should	Local practice can continue where technically possible and sync later.
NFR-007	The system shall support backup and recovery.	Must	Production data can be restored according to documented recovery objectives.

17. System Architecture Specification

Recommended initial architecture: modular client-server architecture with a web application front end, API backend, audio processing module, relational database, object storage for recordings, authentication provider integration, and analytics/reporting services.

Component	Purpose	Recommended Technology Options
Client Application	Practice interface, dashboards, portals, reports	React, Next.js, Electron, React Native, or PWA
API Backend	Business logic, authentication enforcement, user management, reporting	Python FastAPI, Node.js NestJS, or Django
Audio Engine	Pitch detection, note mapping, rhythm analysis, input calibration	Python audio stack, Web Audio API, aubio, librosa for offline analysis
Database	Users, roles, classes, exercises, sessions, scores, policies	PostgreSQL recommended
Object Storage	Audio recordings and generated report files	AWS S3, Azure Blob Storage, Google Cloud Storage, or compatible storage
Authentication	Email/password, OAuth, SSO	Auth0, Azure AD, Google Identity, Microsoft Entra, or custom auth
Reporting Service	PDF and CSV report generation	Server-side report generator with queued jobs for large exports
AI Lab Service	Optional AI feedback and experimentation	Isolated service with strict data usage controls
Notification Service	Emails, reminders, parent alerts, teacher responses	Transactional email provider and internal notification table

18. Database Design Specification

The following logical schema should be used as the starting point for database implementation. Field names may be adapted to the selected framework, but entity relationships should remain intact.

Entity	Key Fields	Purpose
users	id, email, password_hash, display_name, status, created_at, updated_at	Base account identity for every user.
roles	id, name	Student, Teacher, Parent, Institution Admin, System Admin.
user_roles	user_id, role_id, institution_id	Maps users to roles and institutions.
institutions	id, name, type, subscription_status, created_at	Schools, colleges, studios, or organizations.
classes	id, institution_id, teacher_id, name, term, max_students	Teacher-managed instructional groups.
class_students	class_id, student_id, status	Student enrollment in classes.
parent_student_links	parent_id, student_id, relationship, status	Authorizes parent access to student data.
student_profiles	student_id, student_identifier, grade, school, avatar_id	Optional student-specific profile fields.
exercises	id, title, mode, target_note, bpm, time_signature, difficulty, created_by	Practice assignments or library exercises.
assignments	id, class_id, exercise_id, due_date, instructions	Teacher-assigned practice work.
practice_sessions	id, student_id, assignment_id, started_at, ended_at, duration_seconds, mode	Top-level record for a practice session.
pitch_events	id, session_id, timestamp_ms, detected_frequency, detected_note,	Fine-grained pitch analysis events.

	target_note, cents_deviation	
rhythm_events	id, session_id, timestamp_ms, expected_beat_ms, actual_beat_ms, deviation_ms	Fine-grained rhythm analysis events.
recordings	id, session_id, storage_uri, duration_seconds, format, created_at	Audio recording references and metadata.
scores	id, session_id, pitch_accuracy, timing_accuracy, overall_score, computed_at	Computed performance metrics.
reports	id, owner_id, report_type, file_uri, format, created_at	Generated PDF or CSV reports.
questions	id, student_id, teacher_id, class_id, subject, message, status, created_at	Student question workflow.
notifications	id, user_id, type, message, read_at, created_at	Internal notifications and alerts.
ai_settings	id, institution_id, class_id, enabled, data_usage_allowed, updated_by	AI feature controls and policy state.
audit_logs	id, actor_user_id, action, entity_type, entity_id, created_at	Security and administrative audit trail.

19. API Specification

Endpoint	Purpose	Access	Response
POST /auth/register	Create user account	Public	User profile and verification response
POST /auth/login	Authenticate user	Public	Access token and refresh token
GET /me	Get current user profile	Authenticated	User, roles, institutions
GET /classes	List classes visible to user	Teacher/Admin	Class list
POST /classes	Create class	Teacher/Admin	Created class
POST /classes/{id}/students	Add student to class	Teacher/Admin	Enrollment record
GET /exercises	List exercises	Authenticated	Exercise library
POST /assignments	Create assignment	Teacher	Assignment
POST /practice-sessions	Start or save practice session	Student	Session record
POST /practice-sessions/{id}/events	Upload pitch or rhythm events	Student	Accepted event count
POST /recordings	Upload recording metadata and file reference	Student	Recording record
GET /students/{id}/analytics	View student analytics	Authorized Teacher/Parent/Admin/Student	Metrics and trends
GET /reports	List available reports	Authenticated	Report list
POST /reports	Generate report	Authorized roles	Report job or file
POST /questions	Submit question	Student	Question record
POST /questions/{id}/reply	Reply to question	Teacher	Reply confirmation
GET /admin/institution-summary	Institution dashboard	Institution Admin	Usage and progress summary
PUT /ai-settings	Update AI policy settings	Teacher/Admin	Updated AI settings

20. User Interface Specification

Screen	Primary User	Required Elements
Login/Register	All users	Email/password login, OAuth buttons, password reset, account verification
Student Dashboard	Student	Today practice status, assignments, recent accuracy, achievements, start practice button
Practice Screen	Student	Microphone input, calibration, target note, detected note, tuning indicator, metronome, record button
Playback Screen	Student/Teacher	Recording playback, pitch timeline, rhythm timeline, comments, comparison
Teacher Dashboard	Teacher	Class list, alerts, practice compliance, student progress, report generation
Class Detail	Teacher	Roster up to 30 baseline students, assignments, messages, analytics
Assignment Builder	Teacher	Exercise selection, target note, BPM, time signature, due date, instructions
Parent Dashboard	Parent	Linked student progress, practice streak, achievements, notifications
Institution Admin Dashboard	Institution Admin	Users, teachers, classes, licenses, usage reports
AI Lab	Teacher/Admin	AI ON/OFF indicator, controls, policy flags, feedback preview
Reports	Authorized roles	Filters, date ranges, export to PDF and CSV
System Admin Console	System Admin	Licensing, configuration, support tools, audit logs

21. Testing and Acceptance Criteria

Test Category	Purpose	Acceptance Standard
Unit Tests	Validate functions such as note mapping, cents deviation, score calculation, permissions	All critical calculation and permission functions pass.
Integration Tests	Validate API, database, storage, authentication, and reporting flows	Core end-to-end workflows pass in staging.
Audio Accuracy Tests	Validate pitch detection against known tones and recordings	Detected notes match test tones within defined tolerance.
Latency Tests	Validate live feedback responsiveness	Practice screen meets prototype latency target.
Security Tests	Validate authentication, authorization, and data isolation	Users cannot access unauthorized records.
Accessibility Tests	Validate keyboard navigation, labels, contrast, and screen reader basics	Core workflows are usable with assistive tools.
Load Tests	Validate class and institution usage	A class of 30 students remains functional under expected activity.
User Acceptance Tests	Validate teacher, student, parent, and administrator workflows	Representative users complete required tasks without developer

		intervention.
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22. MVP Scope

The first build should focus on a usable minimum viable product before attempting the full institutional platform.

MVP Feature	Included in First Build
User registration/login	Yes
Student dashboard	Yes
Real-time microphone pitch detection	Yes
Detected note and target note display	Yes
Metronome	Yes
Basic recording and playback	Yes
Practice duration tracking	Yes
Teacher class creation	Yes
Assignments	Yes, basic version
Student progress analytics	Yes, basic metrics
PDF/CSV reports	Basic CSV first, PDF next
Parent portal	Phase 2 unless needed immediately
Institution licensing	Phase 2
AI Lab	Phase 3
Mobile apps	Phase 2 or 3 after web/desktop validation

23. Development Backlog

Epic ID	Epic	Description
EPIC-001	Project setup	Repository, environment configuration, coding standards, CI pipeline
EPIC-002	Authentication	Registration, login, password reset, roles, session management
EPIC-003	Audio prototype	Microphone access, calibration, pitch detection, note mapping
EPIC-004	Practice workflow	Start session, target note, feedback, metronome, save results
EPIC-005	Database implementation	PostgreSQL schema, migrations, seed data
EPIC-006	Teacher portal	Class creation, roster, assignment builder, student analytics
EPIC-007	Recording workflow	Record, store, replay, attach to session
EPIC-008	Reports	CSV export, PDF generation, filters, date ranges
EPIC-009	Parent portal	Linked access, progress view, notifications
EPIC-010	Institution admin	Licenses, departments, users, reports
EPIC-011	AI Lab	Policy controls, visible ON/OFF state, feedback experiments

EPIC-012	Security hardening	Audit logs, authorization tests, encryption review
EPIC-013	Deployment	Staging, production, backups, monitoring, documentation

24. Risks, Assumptions, and Open Decisions

Type	Item	Mitigation or Decision Needed
Risk	Real-time pitch detection may vary by microphone quality and room noise.	Add calibration, noise reduction, test with common devices.
Risk	Mobile browser audio permissions can be inconsistent.	Prototype early on target devices.
Risk	Student privacy laws may apply in school deployments.	Perform legal and compliance review before institutional release.
Risk	AI feedback can be inaccurate or discouraging if uncontrolled.	Keep AI Lab optional, teacher-controlled, and clearly labeled.
Assumption	Initial class size baseline is 30 students.	Validate with expected school use cases.
Assumption	15 minutes is the default daily practice target.	Allow institution or teacher override later.
Open Decision	Primary technology stack.	Recommended: React front end, FastAPI backend, PostgreSQL database.
Open Decision	First platform.	Recommended: web application first, then desktop wrapper or mobile as needed.
Open Decision	Cloud provider.	Select AWS, Azure, Google Cloud, or compatible provider based on cost and support needs.

25. Codex-Assisted Implementation Prompt Pack

The following prompts can be used sequentially with a coding assistant to begin implementation.

1. Create a monorepo for Project Pitch Perfect with a React front end, FastAPI backend, PostgreSQL database, Docker Compose development environment, and pytest test structure.
2. Implement the database schema from the SRS using SQLAlchemy models and Alembic migrations.
3. Create role-based authentication with email/password login, password hashing, JWT access tokens, refresh tokens, and role checks for Student, Teacher, Parent, Institution Admin, and System Admin.
4. Build a student practice screen using browser microphone input, real-time pitch detection, target note display, detected note display, and cents deviation feedback.
5. Implement a metronome component with BPM, time signature, audio click, and visual beat indicator.
6. Create API endpoints for classes, class students, exercises, assignments, practice sessions, pitch events, rhythm events, recordings, and analytics.

7. Build a teacher dashboard with class roster, assignment creation, student progress metrics, and CSV export.
8. Build a parent dashboard that shows only linked student progress, practice duration, achievements, and notifications.
9. Add audit logging for administrative actions and role changes.
10. Write unit tests and integration tests for authentication, note mapping, practice session saving, role-based access, and report export.

26. Glossary

Term	Definition
Pitch	The perceived frequency of a musical sound.
Frequency	The measured vibration rate of sound, expressed in Hertz.
Cents deviation	A musical measurement showing how far a detected pitch is from the target pitch.
BPM	Beats per minute.
Time signature	A notation indicating how many beats are in each measure and what note value receives one beat.
Practice session	A recorded or tracked period during which a student practices an exercise.
Assignment	A teacher-created exercise or task assigned to a student or class.
AI Lab	An optional controlled area for AI-assisted feedback and experimental analysis.
RBAC	Role-based access control.
SSO	Single sign-on through an institution identity provider.

27. Appendix A: Recommended First Prototype

The first prototype should prove the hardest technical risk: accurate and responsive pitch detection in a usable practice screen. The prototype should include microphone calibration, note detection, target note, visual tuning indicator, metronome, and local session summary.

28. Appendix B: Implementation Priority Order

11. Repository and environment setup.
12. Authentication and roles.
13. Audio input and pitch detection prototype.
14. Student practice screen.
15. Practice session persistence.
16. Metronome and rhythm support.
17. Teacher class and assignment tools.
18. Analytics and reports.
19. Recording storage.

20. Parent and institution portals.

21. AI Lab and advanced features.